CSE 270 Sports Analytics

Quiz 1

Name: _____

Problem 1

What is the correct way of installing a library in R?

```
A) install.packages("devtools")
B) install.packages(devtools)
C) library("devtools")
D) library(devtools)
```

Problem 2

Do these two codes perform the same action ?

```
a <- c(1:10)
a <- c(1,10)

A) TRUE
B) FALSE
```

Problem 3

```
class_schedules <- c("MWF 12:30-13:20","TT 9:00-10:15", "MWF 08:30-09:20", "MWF 10:30-11:20")
names(class_schedules) <- c("CSE270", "CS222", "CS112", "CS246")
```

a) Get the schedule for CSE 270,
Pick all the correct options.

```
A) class_schedules["CSE270"]
B) class_schedules[0]
C) class_schedules[1]
D) class_schedules$CSE_270
```

b) Get the schedule for all classes except ***CSE 270***,
Pick all the correct options.

```
A) class_schedules[-c("CSE270")]
B) class_schedules[2:4]
C) class_schedules[c(2,4)]
D) class_schedules[names(class_schedules) != "CSE270"]
```

Problem 4

Which of these code chunks will create a 3x3 matrix
and fill it with numbers from 1 to 9.
It should look like this.

```
##      [,1] [,2] [,3]
## [1,]    1    2    3
## [2,]    4    5    6
## [3,]    7    8    9
```

Pick all that apply

```
A) one_to_nine <- matrix(1:9)
B) one_to_nine <- matrix(1:9, byrow=TRUE, nrow=3)
C) one_to_nine <- matrix(c(1,9), byrow=TRUE, nrow=3)
D) one_to_nine <- matrix(rbind(c(1:3), c(4:6), c(7:9)))
```

Problem 5

a) Write down the commands to change the column names and row names of
the matrix ***one__to__nine***, set the column names to ("Lazio", "Roma" , "Juventus") and
the row names to ("2022", "2021", "2020")

```
# Write down your code here, should not take more than two lines
#
#
```

b) The matrix displays the standings for the teams during the last three years, write a command to get Lazio's final position in 2020, can you get it in multiple ways ?

```
# Write down your code here, there are at least 3 ways to do that
#
#
#
```

c) Write down the output of this code chunk

```
rowSums(one_to_nine)
```

```
# The output should not be more than 2 lines
#
#
```

Problem 6

a) Write a code chunk to convert the matrix ***one__to__nine*** into a dataframe and name that dataframe ***standings__2020__2022***.

```
# Write down your code here, should take no more than 1 line
#
```

b) Write a code chunk to add one more column to the data frame, name it "Napoli" and add values (4, 2, 3) for the values of that column.

```
# Write down your code here, should take no more than 2 lines
#
#
```

c) Write a code chunk to add one more row to the data frame, name it 2019 and add the values, (3,4,2,5)

```
# Write down your code here, should take no more than 1 line
#
```

d) Calculate the 1st, 2nd, 3d quartiles abd mean values for each team's standing from 2019-2022

Lazio: 1st Qu:_____ 2nd Qu:_____ 3rd Qu:_____ Mean:_____

Roma: 1st Qu:_____ 2nd Qu:_____ 3rd Qu:_____ Mean:_____

Juventus: 1st Qu:_____ 2nd Qu:_____ 3rd Qu:_____ Mean:_____

Napoli: 1st Qu:_____ 2nd Qu:_____ 3rd Qu:_____ Mean:_____

Problem 7

a) R has a built in dataframe called ***iris*** that contains some information about Species.
Write down the commands to get the data frame and show the last 3 observations of the data frame.
The output should be like this

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
## 148          6.5         3.0          5.2         2.0 virginica
## 149          6.2         3.4          5.4         2.3 virginica
## 150          5.9         3.0          5.1         1.8 virginica
```

```
# Write down your code here, should take no more than 2 lines
#
#
```

b) Write a code chunk to create 2 new variables for the dataframe iris and call them ***Sepal.Area*** and ***Petal.Area*** and fill in the values with the area(width x length) of both variables correspondingly,

```
# Write down your code here, should take no more than 2 lines
#
#
```

c) Write a code chunk to filter out the data frame ***iris*** to only include the species for which the variable ***Sepal.Area*** is greater than 10 and the variable ***Petal.Area*** is less than 1.

```
# Write down your code here, should take no more than 1 line
#
```